Sparse Autoencoders Share a Reproducible Core

Anonymous Author(s)
Affiliation
Address
email

Abstract

Sparse autoencoders (SAEs) are the dominant approach for decomposing neural network activations into interpretable features. Recent work found that SAE features differ substantially across random seeds [**?**], calling into question whether SAE features reflect real model structure. We trace this instability to a single cause: the first principal component of the residual stream explains 93–98% of activation variance in middle layers, and no unique sparse decomposition of this signal exists, so different seeds find different solutions. These seed-dependent features dominate importance rankings, and as a result, the most important features are the least reproducible. Subtracting this component before training reverses the relationship: the top-50 sharing rate jumps from 4% to 86%. After correction, importance predicts reproducibility, and the reproducible features are causally faithful, producing $10\times$ more consistent steering effects across seeds than non-reproducible ones. We verify this across 9 models from 5 families, from 124M to 70B parameters. Our results indicate that SAE instability is dominated by a single removable confound, and that after correction, important SAE features are reliably recoverable across training runs.

1 Introduction

Large language models are capable but opaque: it is not obvious from the weights how a model arrives at its outputs. Mechanistic interpretability aims to close this gap by identifying the internal features an LLM uses in its computations.

Sparse autoencoders (SAEs) are the leading tool for this goal. An SAE is a shallow autoencoder trained to reconstruct a model's internal activations through a bottleneck with a sparsity constraint, decomposing each activation into a sparse combination of learned directions. In principle, each direction corresponds to a feature the model has learned.

For SAE features to be useful, they cannot be artifacts of the SAE's training procedure—they must reflect the model's actual computations. Recent work calls this into question. **?** trained SAEs with different random seeds on the same model and data and found that $\sim$70% of features differ across seeds. This result has contributed to growing skepticism about SAEs, alongside negative downstream evaluations [**??**] and the deprioritization of SAE research by the DeepMind mechanistic interpretability team [**?**].

Despite these concerns, SAEs remain the leading method for unsupervised feature discovery in neural networks, making the question of whether their features are real—and reproducible—worth resolving. We approach this question differently: rather than measuring reproducibility across the full dictionary, we ask whether the features that matter most to the model's predictions are stable across seeds. If a feature corresponds to a real direction in the model's computation, different SAEs should find it regardless of initialization; in other words, the most important features should be the most reproducible. We test this directly by zeroing each feature's activation and measuring the increase in cross-entropy loss, then checking whether the most important features are shared across seeds.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Figure 1: The importance–reproducibility relationship before and after PC1 subtraction. Each point is one SAE feature; the x-axis is ablation importance and the y-axis is cross-seed decoder cosine similarity. **Left:** without correction, the most important features are the least reproducible. **Right:** after subtracting one principal component, the relationship reverses.

They are not. The top-50 most important features are only 4–8% shared, far below the dictionary-wide average of $\sim 50\%$. Ablation importance selects the least reproducible features, not the most (Figure 1, left).

We trace this paradox to a single cause: residual stream anisotropy. Contextual representations in neural language models are known to cluster along a narrow set of directions rather than spreading uniformly [**???**]. We find that in middle layers, a single principal component (PC1) explains 93–98% of activation variance. The SAE dedicates hundreds of features to reconstructing this signal. But no unique sparse decomposition of a continuous signal exists, so different seeds find different solutions. These features score highest on ablation importance because removing any one creates out-of-distribution activation magnitudes downstream.

The fix is simple: subtract PC1 before SAE training. After this correction, the top-50 sharing rate jumps from 4% to 86% (Figure 1, right), and the relationship between importance and reproducibility reverses: the more important a feature, the more reproducible it is. We verify this across 9 models from 5 families (124M–70B parameters), two datasets, and multiple layers. We further show that the reproducible features are causally faithful, producing $10\times$ more consistent steering effects and $4\times$ more accurate causal predictions than non-reproducible features.

2 Related Work

SAEs were first introduced by **?** and **?**, who showed they could recover interpretable features from language model activations. **?** scaled the approach to GPT-4 and introduced the TopK activation function, and **?** applied SAEs to Claude 3 Sonnet. These successes led to optimism that SAEs could, eventually, be used to fully enumerate the features a model uses, enabling safety auditing and circuit-level understanding [**?**].

This enthusiasm rests on a theoretical prediction: **?** showed in toy models of superposition that more important features receive dedicated linear directions, while less important ones are compressed

into shared subspaces. An SAE searches for exactly these directions — sparse linear combinations that reconstruct the activation. If this prediction holds in real models, the most important features should be the easiest to find, and any SAE should find them, regardless of initialization. Recent work suggests otherwise [**????**].

Our explanation draws on a geometric property of the residual stream that has been studied extensively outside the interpretability literature. **?** showed that contextual representations in BERT, ELMo, and GPT-2 cluster in a narrow cone rather than spreading uniformly. **?** proposed a fix: removing the mean and top principal components from embeddings ("All-but-the-Top"), which improved isotropy and downstream task performance. **?** observed that decoder-only transformers exhibit a bell-shaped anisotropy profile, peaking in middle layers and declining toward the output. **?** argued this pattern is inherent to self-attention, not merely a consequence of the training objective. Our contribution connects this literature to SAE evaluation: the same anisotropy that distorts embedding geometry also distorts SAE importance rankings.

The features at the center of our analysis correspond to what **?** call "dense latents," or SAE features that fire on nearly all tokens. Sun et al. showed these often reflect real model structure, but noted that dense latents may learn a basis that spans the true dense subspace without uniquely aligning with it, because linear combinations reconstruct equally well. We identify what this subspace is: the dominant direction in the residual stream (PC1), along which tokens vary primarily in magnitude. The non-identifiability Sun et al. observed is the mechanism behind the importance–reproducibility paradox we report in Section 4. Sun et al. identify dense latents by firing frequency; we show this criterion is insufficient and that geometric identification via PC1 alignment is necessary (Section **??**).

3 Experimental Setup

We train TopK sparse autoencoders (k=64, dictionary size $4\times$ hidden dimension) on residual stream activations at the middle layer of each model, following **?**. We evaluate on 9 models from 5 families: Pythia-160M, 1.4B, and 6.9B; GPT-2-small; Gemma 2 2B and 9B; Llama 3.1 8B and 70B; and Qwen 2.5 7B. For each model, we train two SAEs on 5M tokens from WikiText-103 that differ only in their random seed. Training details are in Appendix **??**.

We measure each feature's *causal importance* via zero-ablation: we insert the SAE at the target layer, set the feature's activation to zero, and record the increase in cross-entropy loss. We measure *reproducibility* by computing the maximum cosine similarity between each feature's decoder direction and all decoder directions in the second SAE [**?**]. A feature is *shared* if this maximum exceeds 0.7, following **?**; otherwise it is an *orphan*. Results hold across thresholds from 0.5 to 0.95 (Appendix **??**).

Our primary question is: among the top-k features ranked by importance, what fraction are shared?

4 Results

Our results proceed in four parts. First, we show that the most important SAE features are the least reproducible across seeds — a paradox, given that important features should be the easiest to find (Section 4.1). Second, we explain why: residual stream anisotropy creates a population of features that dominate importance rankings but admit no unique sparse decomposition (Section 4.2). Third, we show that subtracting a single principal component before training reverses the relationship, making importance predict reproducibility and obtaining highly interpretable, monosemantic features (Section 4.3). Fourth, we verify the effect generalizes across 8 models from 5 families and rule out alternative explanations (Section 4.4).

4.1 Important features are not reproducible

We rank features by ablation importance and check whether the most important features are shared across seeds. Table 1 reports top-50 sharing rates for each model, ordered by PC1 dominance.

In every model with PC1 above 93%, the top-50 most important features are only 4–10% shared, compared to dictionary-wide baselines of 46–62%. Llama 3.1 8B is a partial exception (46%), possibly reflecting architectural differences in how Llama distributes computation. In Gemma 2 9B,

Table 1: Top-50 sharing rates versus dictionary-wide baseline. In models with high PC1 dominance, the most important features are far less reproducible than average. In Gemma models, where PC1 explains less variance, this inversion is absent.

Model	Family	PC1 (%)	Baseline shared (%)	Top-50 shared (%)
Qwen 2.5 7B	Qwen	99.9	47	8
GPT-2-small	GPT-2	97.0	46	6
Llama 3.1 8B	Llama	96.9	61	46
Pythia-1.4B	Pythia	96.4	62	10
Pythia-6.9B	Pythia	95.7	62	6
Pythia-160M	Pythia	93.4	49	4
Gemma 2 2B	Gemma	78.2	60	70
Gemma 2 9B	Gemma	69.9	57	86

Table 2: The top features by importance at k=0 in Qwen 2.5 7B. Confounding features (top) fire on nearly all tokens with identical token distributions. The only shared features (bottom) fire on specific tokens and are the only features in the top 50 with low PC1 alignment.

Feature	Imp.	Firing	PC1 $\cos^2$	Shared	Top tokens
14126	0.537	98.2%	0.260	×	*the*, , . *of and in* . . .
9091	0.522	98.2%	0.229	×	*the*, , . *of and in* . . .
9651	0.462	98.2%	0.243	×	*the*, , . *of and in* . . .
9700	0.440	98.2%	0.249	×	*the*, , . *of and in* . . .
516	0.188	3.0%	0.001	✓	spaces before numbers
3974	0.112	5.7%	0.003	✓	*The*, *the*
2094	0.061	3.6%	0.000	✓	sentence-ending periods

where PC1 explains only 70% of variance, the pattern reverses: the top-50 sharing rate is 86%, above baseline.

What are these important-but-unreproducible features? Table 2 shows representative features from the top 50 in Qwen 2.5 7B.

The confounding features fire on 98.2% of all tokens, with importance values 3–5× higher than any shared feature. They activate on the same tokens — "the", commas, periods, "of", "and" — not because they encode punctuation or function words, but because they fire on everything. The 4 shared features that survive in the top 50 are the only ones not aligned with PC1: they fire on specific, identifiable token classes.

At Pythia-1.4B, five of the top six confounding features fire on the exact same 773 token positions out of 49,382 — pairwise Jaccard overlap of 1.0000, with activation magnitudes correlated at $r > 0.94$. For context, two random features firing at the same 1.6% rate would share approximately 12 positions. Five dictionary entries are reconstructing one signal; a different seed decomposes it differently, which is exactly what we observe.

Across models, these confounding features constitute a small fraction of the dictionary but dominate the importance ranking (Table 3).

In GPT-2-small, 14% of features account for 90% of total ablation importance. In Qwen, 3% account for 89%. A small population of features dominates the importance ranking, and these are precisely the features that differ across seeds. Section 4.2 identifies the underlying signal they reconstruct.

4.2 Residual stream anisotropy explains the paradox

The residual stream in transformer middle layers is dominated by a single direction. Prior work has documented this anisotropy in contextual representations [**??**]. Across the 9 models we evaluate, PC1 of the residual stream explains 70–99.9% of the total variance, with models above 1B parameters typically exceeding 95%. The dominance follows a bell-shaped profile across layers — at Pythia-160M, PC1 explains 5.4% at the embedding layer, rises to 93.4% by layer 6, and drops to 6.5% at the

Table 3: Dense features (decoder $\cos^2$ with PC1 $> 5\times$ median) account for a small fraction of the alive dictionary but dominate total ablation importance.

Model	% of dictionary	% of importance
GPT-2-small	14.0	89.5
Qwen 2.5 7B	2.9	88.5
Pythia-160M	15.7	86.1
Pythia-1.4B	7.8	81.5
Pythia-6.9B	1.3	64.3
Llama 3.1 8B	8.1	53.6
Gemma 2 9B	3.5	51.8
Gemma 2 2B	8.9	49.0

final layer (Appendix **??**). An untrained Pythia-160M shows PC1 $\approx 5\%$ throughout; training creates the anisotropy.

The SAE must reconstruct its input, and PC1 is most of the input. Given a centered activation $\mathbf{x}$, the SAE approximates:

$$\mathbf{x} \approx \sum_i a_i \mathbf{d}_i \tag{1}$$

where a_i are sparse activations and $\mathbf{d}_i$ are unit-norm decoder directions. The optimal reconstruction dedicates significant capacity to the PC1 component $(\mathbf{x}^\top \mathbf{v}_1)\mathbf{v}_1$, which carries $>93\%$ of $\|\mathbf{x}\|^2$. But a sparsity constraint forces the SAE to represent this continuous, one-dimensional signal using a discrete set of features. An overcomplete dictionary ($4\times$ the model dimension) contains far more directions than the space has dimensions, so many sparse subsets can approximate any given direction equally well — the decomposition is not unique [**?**]. This is why features 1482, 7290, 4678, 5331, and 674 in Pythia-1.4B fire on the same 773 positions with correlated magnitudes: they are five ways of tiling one direction.

Dense features from two independently trained SAEs on Pythia-1.4B span overlapping subspaces: principal angle overlap of 0.758, compared to 0.030 for a random baseline ($25\times$). PCA on the same activations recovers the subspace with overlap 0.943. Both SAEs find the same structure; the sparsity constraint forces them to express it in different, seed-dependent coordinates.

These features dominate importance because zeroing one removes part of the SAE's reconstruction of PC1, creating out-of-distribution activation magnitudes downstream. Group ablation on Pythia-1.4B confirms redundancy: individual importance sums to $2.18\times$ the joint effect.

If this explanation is correct, removing PC1 before SAE training should eliminate the confound.

4.3 PC1 subtraction reverses the relationship

We subtract the projection onto PC1 from centered activations before passing them to the SAE:

$$\mathbf{x}_{\text{residual}} = (\mathbf{x} - \boldsymbol{\mu}) - ((\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{v}_1)\, \mathbf{v}_1 \tag{2}$$

During inference, PC1 passes through the model untouched and is restored before the output continues to the next layer — the model sees the same activations it always sees, but the SAE no longer needs to reconstruct the dominant direction.

On Pythia-1.4B, this reverses the relationship completely; the top-50 sharing rate rises from 10% to 90%, and the cross-entropy overhead improves slightly (3.197 to 3.186). Because PC1 is restored exactly during inference, the total reconstruction is determined by the SAE's approximation of the non-PC1 component. The SAE reconstructs this smaller signal more faithfully than it reconstructs the full activation. Removing further principal components adds nothing; the top-50 sharing rate is flat from $k{=}1$ to $k{=}50$ (Appendix **??**).

Table 4 shows the most important features after PC1 bypass on Pythia-1.4B.

Where the confounding features fired on everything and rarely matched across seeds, these features fire on narrow, coherent token classes and are recovered almost identically by an independent SAE. Of the top 50 features after bypass, 48 are shared across seeds (96%). The 2 orphans encode WikiText-specific formatting artifacts. Full feature reports are in Appendix **??**.

Table 4: Top features by importance after PC1 bypass (Pythia-1.4B). Each feature fires on a coherent semantic or syntactic class and is recovered at high cosine similarity by an independently trained SAE. Bolded tokens indicate the triggering position.

Feature	Cosine	Freq	Example contexts
5748	0.989	4.0%	"off the **coast** of Mexico"; "Numerous **ship**wrecks"; "the **seas** around Papa St"
3415	0.961	2.2%	"completed a **flying** instructors course"; "No. 2 **Air** Navigation School"; "There is an **airstr**ip"
7668	0.971	0.2%	"Slayer have **head**lined"; "is **head**quartered at"; "7), **head**quartered at"
5127	0.885	4.8%	"for three years**,** and its"; "circular enclosure**,** as often found"
4110	0.996	38.2%	"built in **1908**"; "On June **18**"; "Boise, **Idaho**"

Table 5: Top-50 sharing rates before and after PC1 bypass. Bypass lift scales with PC1 dominance (Pearson $r = 0.918$; Figure 2).

Model	Family	PC1 (%)	k=0 (%)	k=1 (%)	Lift
Qwen 2.5 7B	Qwen	99.9	8	96	+88
GPT-2-small	GPT-2	97.0	6	76	+70
Llama 3.1 8B	Llama	96.9	46	94	+48
Pythia-1.4B	Pythia	96.4	10	90	+80
Pythia-6.9B	Pythia	95.7	6	90	+84
Pythia-160M	Pythia	93.4	4	78	+74
Gemma 2 2B	Gemma	78.2	70	94	+24
Gemma 2 9B	Gemma	69.9	86	90	+4

4.4 Generalization and controls

Section 4.3 demonstrated the effect on a single model. We now test whether it generalizes, and check the most natural alternative explanations for the effect. Table 5 reports top-50 sharing rates before PC1 bypass (k=0, standard SAE) and after (k=1, PC1 subtracted) across all 8 models.

The bypass lift scales with PC1 dominance (Pearson $r = 0.918$; Figure 2). At Qwen 2.5 7B, PC1 explains 99.9% of variance. The confound is so strong that there is zero overlap between the top-50 features before and after bypass; the SAE finds entirely different features. Top-50 sharing rises from 8% to 96%. At Gemma 2 9B, PC1 explains only 70%. The top features are already semantic and shared without intervention, and the bypass adds only 4 percentage points.

PC1 bypass requires retraining the SAE. A less dramatic intervention is to train a standard SAE and exclude confounding features post-hoc by their decoder alignment with PC1 ($\cos^2 > 5\times$ median). On Pythia-1.4B, this achieves a clean top-50 sharing rate of 88%, compared to 90% with full bypass. The post-hoc method is sufficient for evaluation, though bypass produces slightly cleaner features.

We run two controls. On an untrained Pythia-160M with random weights, PC1 explains only 4.7% of variance and the importance–reproducibility inversion is absent (top-50 sharing = 20% against a 3% baseline), indicating the confound requires learned anisotropy. To test whether removing any set of outlier features would produce the same effect, we exclude 679 features at random — matching the count removed by the PC1 criterion — across 1,000 trials. The mean top-50 sharing is 9% and the maximum is 12%; the PC1 criterion achieves 96%. The improvement is specific to the geometric criterion we propose, not a consequence of excluding outliers in general.

The effect also holds across layers 6, 12, and 18 of Pythia-1.4B (Appendix **??**), across domains when training on OpenWebText instead of WikiText (Appendix **??**), and across matching thresholds from 0.5 to 0.95 (Appendix **??**). Results hold across top-k from 20 to 500 (Appendix **??**).

Finally, a natural alternative to geometric identification is to exclude features by firing frequency, as proposed by **?**. We find that excluding features with firing rates above 3%, 5%, 10%, or 15% produces top-50 sharing of 0–4% — no better than no exclusion. This fails because many confounding

Figure 2: Bypass lift scales with PC1 dominance across 8 models from 5 families. Each point is one model; the x-axis is PC1 variance explained and the y-axis is the change in top-50 sharing rate after PC1 subtraction.

features fire on only 1–5% of tokens individually while aligning collectively with PC1; geometric identification is necessary.

5 Causal Validation

Reproducibility and interpretability alone do not guarantee that a feature reflects real model computation, as interpretability is a human judgment. A stronger test is causal faithfulness: does a feature's decoder direction predict its actual effect on model outputs? We evaluate this on Pythia-1.4B.

Steering. We inject matched feature decoder directions from two independently trained SAEs into the residual stream during generation ($\alpha = 5$) and measure the KL divergence between the resulting output distributions. For shared features, the cross-seed divergence (0.001) is $6\times$ smaller than the steering effect itself (0.006) — both SAEs steer the model to the same place. For orphan features, the cross-seed divergence (0.010) is $2\times$ larger than the steering effect (0.005); the disagreement between SAEs exceeds the effect of steering. Table 6 shows representative examples from three shared features.

Logit lens faithfulness. Each feature's decoder direction, projected through the unembedding, predicts which tokens the feature should promote. We clamp each feature at $+5\sigma$ and measure whether the output distribution shifts toward these predicted tokens. Shared features produce a $4.22\times$ probability increase toward their predicted tokens. Orphan features produce $1.20\times$ ($p < 10^{-10}$, Mann-Whitney U). The decoder directions of shared features predict their causal behavior; orphan features' decoder directions do not.

After PC1 correction, the important features are reproducible, interpretable, and causally faithful. Their decoder directions predict their effects on model outputs, and independently trained SAEs steer the model toward the same concepts.

6 Discussion

7 Conclusion

References

Table 6: Steering with shared features from two independently trained SAEs on Pythia-1.4B ($\alpha = 5$–10). Both SAEs steer the model toward the same domain despite being trained from different random seeds.

Feature	Cosine	Unsteered	Seed 42	Seed 137
5748	0.989	"many parks and gardens, and the city's main park, the Central Park, was a popular destination…"	"many **canals**, which were used for **shipping** and for pleasure. The city was also known for its many **shipyards**…"	"many **shipyards**, and the city was also known for its many **canals**. The city's **fishing** industry was a major source of…"
3736	0.954	"many parks and gardens, and the city's main park, the Central Park, was a popular destination for the rich and famous."	"fine arts, including the famous French Impressionist **painters** Claude **Monet** and Camille **Pissarro**, and for the city's many **opera** and **ballet** companies."	"rich history… The area also has a long history of **art**, most notably in the form of **paintings** and **sculpture**."
3415	0.961	"the first of the two-year-old buildings was opened to the public, although it had not yet been finished. The project was described…"	"the first of the two-year-old **F-15s** was delivered to the **United States Air Force**. The second F-15 was delivered in 1989."	"the first of the two-year-old **Boeing 747-400s** was delivered to the **airline**. The second was delivered in the spring of 1988."